

Git Basics & Common Commands Guide

This guide provides a practical reference for Git operations, ideal for managing projects, handling branches, commits, and resolving conflicts.

1. Setting Up a Git Repository

Initialize a new Git repo:

```
cd /path/to/your/project
```

```
git init
```

Check Repo Status:

```
git status
```

2. Creating and Working with Branches

Create a new branch:

```
git checkout -b branch-name
```

List branches:

```
git branch    # Local branches
```

```
git branch -a # All branches, local + remote
```

Switch branch:

```
git checkout branch-name
```

3. Adding Files and Committing

Stage files:

```
git add file1 file2 # Specific files
```

```
git add .           # All files (be careful!)
```

Commit changes:

```
git commit -m "Your commit message"
```

4. Handling Mistakes

Unstage files:

git reset file-name

Example: git reset .env node_modules

Discard local changes:

git checkout -- file-name

5. Using .gitignore

Create .gitignore:

touch .gitignore

Add files/folders to ignore:

echo ".env" >> .gitignore

echo "node_modules/" >> .gitignore

echo "playwright-report/" >> .gitignore

echo "reports/" >> .gitignore

echo "test-results/" >> .gitignore

6. Pushing Changes to GitHub

Add remote repository:

git remote add origin https://github.com/username/repo.git

Push a branch:

git push -u origin branch-name

Push after commits:

git add .

git commit -m "Commit message"

git push

7. Pulling Changes and Resolving Conflicts

Pull remote changes:

git pull origin branch-name

Resolving merge conflicts:

Git marks conflicts like this:

<<<<<< HEAD

Your changes

=====

Remote changes

>>>>>> branch-name

Steps to resolve:

1. Edit the file manually and remove conflict markers.
2. Stage the resolved file: `git add file-name`
3. Commit the merge: `git commit -m "Resolved merge conflict in file-name"`

8. Viewing Git History

`git log`

`git log --oneline --graph --all`

9. Quick Reference Cheat Sheet

Task	Command
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Initialize repo	<code>git init</code>
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Check status	<code>git status</code>
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Add files	<code>git add file1</code> or <code>git add .</code>
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Commit changes	<code>git commit -m "message"</code>
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Create branch	<code>git checkout -b branch-name</code>
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Switch branch	<code>git checkout branch-name</code>
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List branches	<code>git branch</code> / <code>git branch -a</code>
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Push branch	<code>git push -u origin branch-name</code>
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Pull changes	<code>git pull origin branch-name</code>
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Unstage file	<code>git reset file-name</code>
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Ignore files	<code>.gitignore</code>
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Resolve conflicts	Edit -> <code>git add</code> -> <code>git commit</code>
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